Section A: 12-Week Technical Roadmap

This roadmap outlines the strategic execution plan to deliver a resilient, scalable, and observable multi-country integration platform. The plan is divided into three parallel workstreams to ensure consistent progress across all technical pillars.

Summary Table

Phase	Weeks	Reliability	Integration Modernization	Observability / Ops
1 — Foundation & Standardization	1–4	Idempotency & Retry standards; Common Error Handling framework	C4E initialization; System APIs for Salesforce	Correlation ID policy; Base OTel collector infrastructure
2 — Implementation & Resilience Scaling	5–8	Circuit Breakers & Bulkheads; Anypoint MQ for async flows	Multi-country Process APIs; Legacy integration migration	OTel instrumentation across all layers; Real-time health dashboards
3 — Optimization & Full Visibility	9–12	Chaos Engineering drills; Cache fine-tuning	Country-specific Experience APIs; Developer Portal on Anypoint Exchange	Proactive alerting & SLOs; Post-implementation review

---

Critical Cross-Workstream Dependencies

Before detailing each phase, the following dependencies are critical to execution sequencing:

- The **OTel Collector** (Observability, Weeks 1–4) is a prerequisite for full-layer instrumentation (Observability, Weeks 5–8).
- **Salesforce System APIs** (Integration, Weeks 1–4) are a prerequisite for **Process API** development (Integration, Weeks 5–8).
- **Circuit Breakers and Bulkheads** (Reliability, Weeks 5–8) must be active before deploying **Experience APIs** (Integration, Weeks 9–12).

- **Real-time health dashboards** (Observability, Weeks 5–8) are a prerequisite for **Chaos Engineering drills** (Reliability, Weeks 9–12) — chaos cannot be executed without observability in place.

Roles and Responsibilities

Each workstream has a primary owning team, though cross-team collaboration is continuous throughout:

- **Reliability**: Platform Team / MuleSoft Architects — responsible for resilience patterns, RTF configuration, and integration standards.
- **Integration Modernization**: Integration Team / API Developers — responsible for designing and developing APIs across all three layers (Experience, Process, System).
- **Observability / Ops**: DevOps / SRE Team — responsible for OTel infrastructure, dashboards, alerting, and chaos drill execution.

Phase 1: Foundation & Standardization (Weeks 1–4)

Goal: Establish core architecture, governance models, and shared libraries.

Reliability

- Define global standards for **Idempotency** and **Retries** with exponential backoff and jitter.
- Develop a reusable **Common Error Handling** framework in MuleSoft to ensure uniform error responses across all country APIs.

Integration Modernization

- Initialize the **C4E (Center for Enablement)** to govern API standards and promote asset reuse.
- Design and develop the **System APIs** for Salesforce to abstract the core data model from downstream consumers.

Observability / Operations

- Define a unified **Correlation ID** policy for end-to-end tracking across the digital channel.
- Set up the base **OpenTelemetry (OTel)** collector infrastructure for centralized metric gathering.

Definition of Done — Phase 1

- 100% of APIs have the Correlation ID policy implemented and propagated.
- Common Error Handling framework documented, published to Anypoint Exchange, and adopted across all teams.
- Salesforce System APIs deployed to the development environment with approved API contracts.
- OTel infrastructure operational and receiving baseline telemetry.

Phase 2: Implementation & Resilience Scaling (Weeks 5–8)

Goal: Deploy core business logic and harden the system against failures.

Reliability

- Implement **Circuit Breakers** and **Bulkheads** for all third-party integrations, including payment gateways and identity verification (KYC) services.
- Configure **Anypoint MQ** for asynchronous processing of mission-critical persistence flows such as policy issuance.

Integration Modernization

- Develop **Process APIs** for multi-country business logic, such as premium calculation and risk assessment.
- Migrate high-priority legacy integrations to the new API-Led layers to reduce technical debt.

Observability / Operations

- Integrate OTel instrumentation into all MuleSoft layers (Experience, Process, System) for full-stack visibility.
- Create real-time health dashboards to monitor regional traffic and error rates.

Definition of Done — Phase 2

- Circuit breakers active and validated across 100% of third-party provider integrations.
- Policy issuance flow processing correctly in asynchronous mode via Anypoint MQ.
- Distributed traces visible end-to-end (Frontend → Salesforce) in the OTel dashboard.
- At least one high-priority legacy integration migrated to the new API-Led model.

Phase 3: Optimization & Full Visibility (Weeks 9–12)

Reliability

- Execute **Chaos Engineering** drills to validate system behavior under failure conditions using a simulated flaky upstream service.
- Fine-tune **caching strategies** (Object Store/Redis) to optimize Salesforce API consumption and reduce latency.

Integration Modernization

- Roll out country-specific **Experience APIs** for the initial target markets (Colombia and Mexico).
- Finalize the **Developer Portal** in Anypoint Exchange for team-owned asset management.

Observability / Operations

- Configure proactive alerting based on **SLOs (Service Level Objectives)** defined per service and region.
- Conduct a post-implementation review documenting lessons learned, residual technical debt, and recommendations for the next iteration.

Definition of Done — Phase 3

- Chaos drills executed with no service degradation in the unaffected region (Bulkhead validation confirmed).
- P99 end-to-end latency below 500ms under normal load conditions.
- Colombia and Mexico Experience APIs deployed to production environment.
- SLOs defined, documented, and active alerts configured in the monitoring dashboard.